Sea-Bird Electronics, Inc.

13431 NE 20th Street, Bellevue, WA 98005-2010 USA

Phone: (+1) 425-643-9866 Fax (+1) 425-643-9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 9171
CALIBRATION DATE: 27-Mar-14

Slocum Payload CTD CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -9.800330e-001                CPcor = -9.5700e-008
h =  1.716853e-001                CTcor =  3.2500e-006
i = -2.785892e-004                WBOTC =  5.5742e-007
j =  5.048565e-005
```

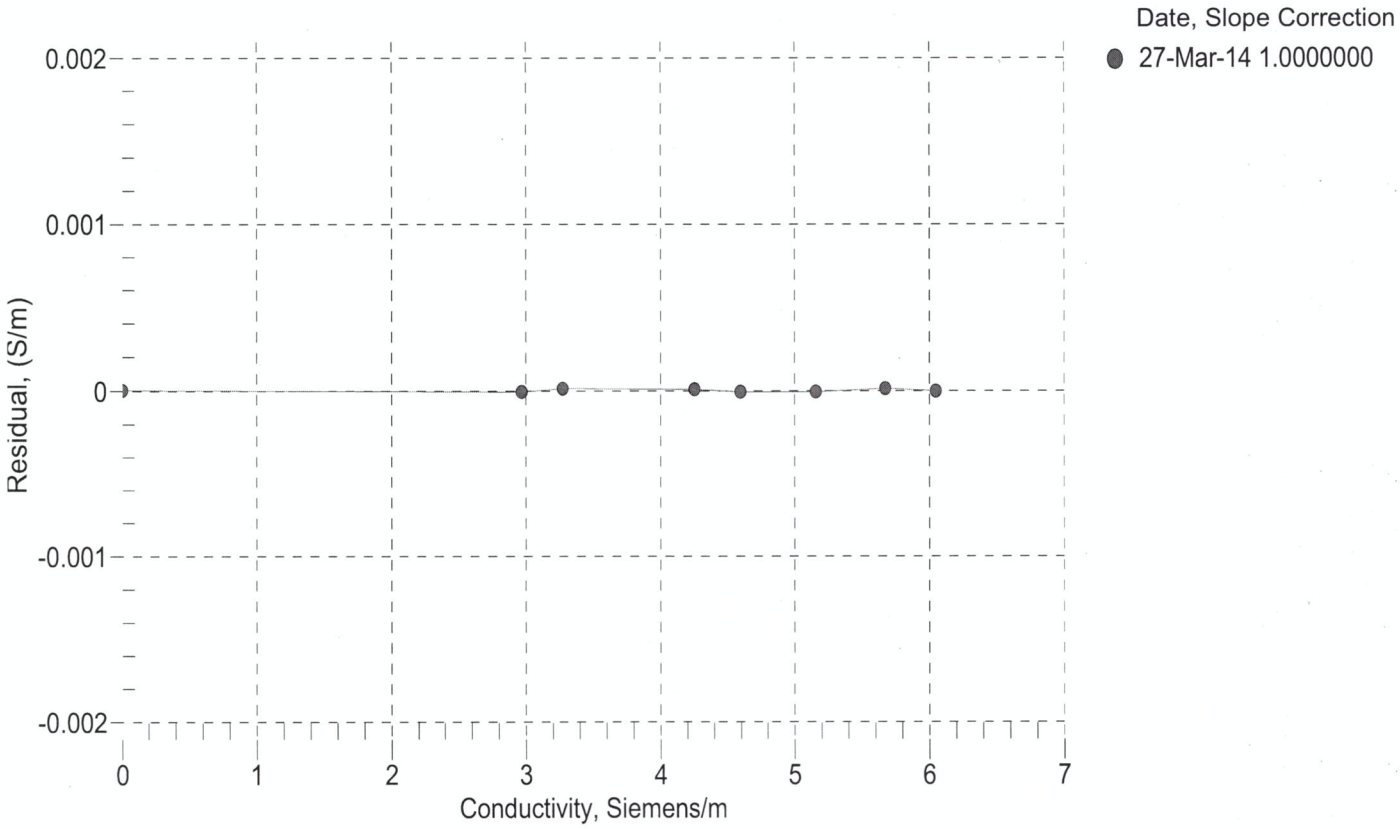
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2391.82	0.00000	0.00000
1.0001	34.7131	2.96799	4797.82	2.96798	-0.00001
4.5000	34.6927	3.27421	4979.87	3.27422	0.00001
15.0000	34.6509	4.25346	5521.23	4.25347	0.00001
18.5000	34.6419	4.59772	5699.09	4.59772	-0.00001
24.0000	34.6321	5.15425	5975.19	5.15424	-0.00001
29.0000	34.6269	5.67478	6222.09	5.67479	0.00001
32.5000	34.6244	6.04630	6392.30	6.04630	-0.00000

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductiv ity = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$ Siemens / meter

t = temperatur e[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 9171
CALIBRATION DATE: 24-Mar-14

Slocum Payload CTD PRESSURE CALIBRATION DATA
FSR: 1450 psia S/N 4128702

COEFFICIENTS:

```
PA0 =       2.007332e-001
PA1 =       4.451630e-003
PA2 =      -1.623311e-011
PTEMPA0 = -6.983828e+001
PTEMPA1 =  4.984380e-002
PTEMPA2 = -1.905604e-007
```

```
PTCA0 =  5.252162e+005
PTCA1 =  1.997792e+000
PTCA2 = -7.056556e-003
PTCB0 =  2.552775e+001
PTCB1 = -1.050000e-003
PTCB2 =  0.000000e+000
```

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	THERMISTOR OUTPUT	COMPUTED PRESSURE	ERROR %FS
14.67	528511.0	1849.0	14.70	0.00
315.03	595933.0	1855.0	315.03	-0.00
615.01	663320.0	1856.0	615.05	0.00
915.04	730744.0	1856.0	915.10	0.00
1215.05	798195.0	1858.0	1215.11	0.00
1464.96	854380.0	1858.0	1464.91	-0.00
1215.06	798186.0	1857.0	1215.07	0.00
915.17	730754.0	1857.0	915.14	-0.00
615.17	663333.0	1859.0	615.11	-0.00
315.18	595948.0	1857.0	315.10	-0.01
14.66	528507.0	1858.0	14.68	0.00

THERMAL CORRECTION

TEMP ITS90	THERMISTOR OUTPUT	INST OUTPUT
32.50	2070	528506.60
29.00	1998	528503.20
24.00	1896	528495.00
18.50	1784	528485.80
15.00	1714	528477.40
4.50	1500	528457.40
1.00	1429	528453.80

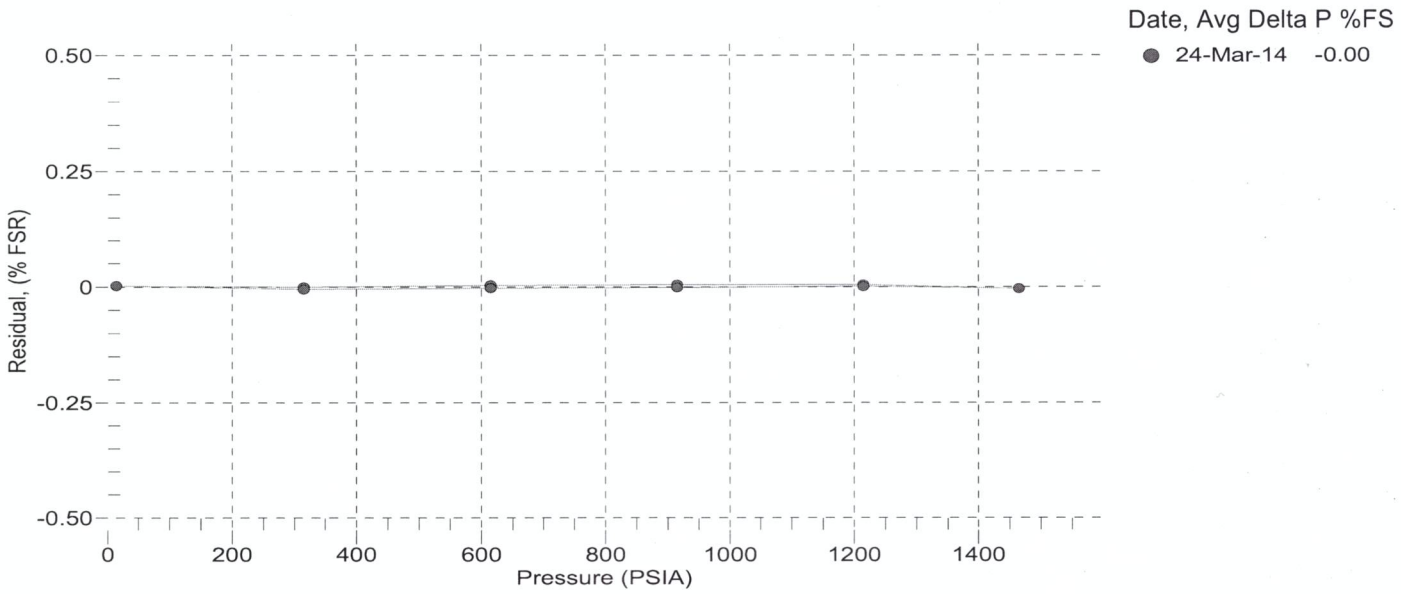
TEMP (ITS90)	SPAN (mV)
-5.00	25.53
35.00	25.49

y = thermistor output; $t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = pressure\ output - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$pressure\ (psia) = PA0 + PA1 * n + PA2 * n^2$



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CALIBRATION DATE: 27-Mar-14

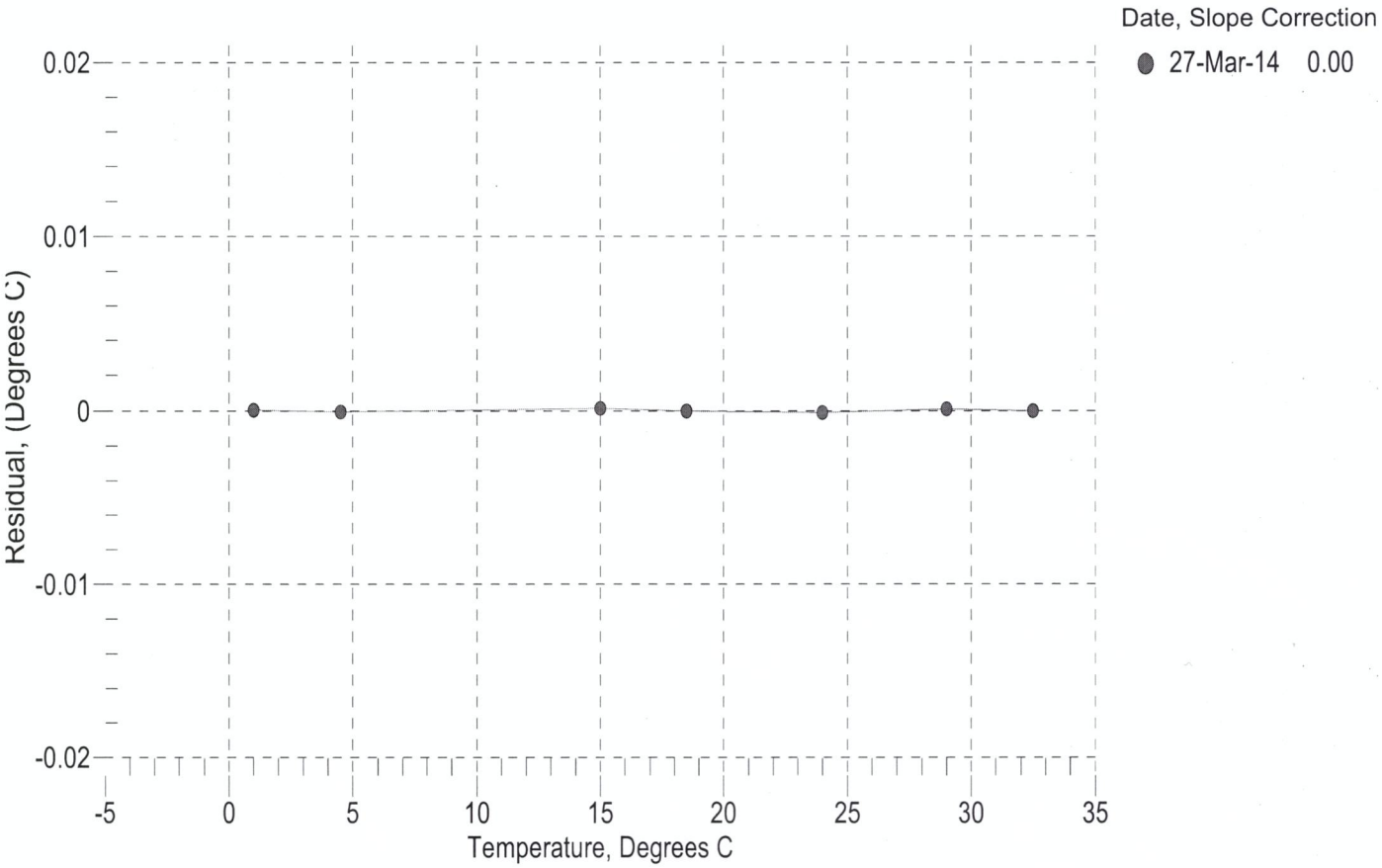
Slocum Payload CTD TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

```
a0 = -5.968851e-005
a1 =  2.997384e-004
a2 = -4.118427e-006
a3 =  1.928339e-007
```

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0001	588368.4	1.0001	0.0000
4.5000	502610.8	4.4999	-0.0001
15.0000	319541.4	15.0001	0.0001
18.5000	276494.0	18.5000	-0.0000
24.0000	221586.6	23.9999	-0.0001
29.0000	182306.8	29.0001	0.0001
32.5000	159567.8	32.5000	-0.0000

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature

n = instrument output